

Lecture 9 Relational Database Design

Functional Dependencies

United International College

Outline

- Features of Good Relational Design
- Atomic Domains and First Normal Form
- **Functional Dependency Theory**
- BCNF
- 3rd Normal Form
- Multivalued Dependencies and Decomposition
- Database-Design Process

Functional Dependencies

- Functional dependencies are some constraints on the set of legal relations.
- The constraint is that the value for a certain set of attributes **uniquely determines** the value for another set of attributes.
- A functional dependency is a **generalization of** the notion of a **key**.

Functional Dependencies - Definition

- Let R be a relation schema, α and β are two attribute sets, and
$$\alpha \subseteq R \text{ and } \beta \subseteq R$$

- The functional dependency

$$\alpha \rightarrow \beta$$

holds on R if and only if for any legal relations $r(R)$, whenever any two tuples t_1 and t_2 of r agree on the attributes α , they also agree on the attributes β . That is,

$$\text{If } t_1[\alpha] = t_2[\alpha], \text{ then } t_1[\beta] = t_2[\beta].$$

- Read as “ α determines β ” or “ β depends on α ”.
- Example: consider $r(A, B)$ with the following instance of r .

A	B
1	4
1	5
3	7

- On this instance, $A \rightarrow B$ does **NOT** hold, but $B \rightarrow A$ does hold.

Example

- Which FDs may exist in this relation $R(A, B, C, D)$?

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>a1</i>	<i>b1</i>	<i>c1</i>	<i>d1</i>
<i>a1</i>	<i>b2</i>	<i>c2</i>	<i>d2</i>
<i>a2</i>	<i>b2</i>	<i>c2</i>	<i>d3</i>
<i>a3</i>	<i>b3</i>	<i>c4</i>	<i>d3</i>

- $B \rightarrow C, C \rightarrow B$
- $AB \rightarrow C$
- The combination of any two attributes (except BC) determines others.

Functional Dependencies - Properties

- K is a superkey for relation schema R if and only if $K \rightarrow R$.
- K is a candidate key for R if and only if
 - $K \rightarrow R$, and
 - for no $\alpha \subsetneq K$, $\alpha \rightarrow R$.
- Functional dependencies allow us to express constraints that cannot be expressed using superkeys. Consider the schema:
 $course_info = (\underline{c_name}, p_code, credits, domain, c_number)$

We expect this functional dependency to hold

$$c_name \rightarrow credits$$

but would not expect the following to hold:

$$credits \rightarrow c_name.$$

Use of Functional Dependencies

- We use functional dependencies to:
 - test relations to see if they are legal under a given set of functional dependencies.
 - If a relation r is legal under a set F of functional dependencies, we say that r **satisfies** F .
 - specify constraints on the set of legal relations
 - We say that F holds on the schema R if all legal relations on R satisfy the set of functional dependencies F .

Use of Functional Dependencies

- Note: A specific instance of a relation schema may satisfy a functional dependency even if the functional dependency does not hold on all legal instances.
- For example, a **specific instance** of *course* may, by chance, satisfy

“If two courses have different *credits*,
their *course names* are different.”

- In consequence, this functional dependency seems hold.
 $credits \rightarrow c_name$

But it doesn't mean “*customer_name* depends on *credits*” is true **in general**.

Functional Dependencies (Cont.)

- A functional dependency is **trivial** if it is satisfied by all instances of a relation.
- Equivalently,

If $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$ is trivial.

Example:

- $credit, domain, c_number \rightarrow c_number$
- $c_name \rightarrow c_name$

Closure of a Set of Functional Dependencies

- Given a set F of functional dependencies, there are certain other functional dependencies that are **logically** implied by F .
- For example:

If $A \rightarrow B$ and $B \rightarrow C$, then we can **infer** that $A \rightarrow C$.

The reason is that by the definition of functional dependencies,

- if two tuples t_1, t_2 have the same value on A , then t_1 and t_2 have the same value on B ;
 - if t_1, t_2 have the same value on B , then they have the same value on C ;
 - thus, if t_1, t_2 have the same value on A , then they have the same value on C ;
 - which is $A \rightarrow C$.
- The set of all functional dependencies logically implied by F is the **closure** of F , denoted by F^+ .
 - F^+ is a superset of F .

Closure of a Set of Functional Dependencies

- We can find all of F^+ by applying Armstrong's Axioms:
 1. **reflexivity**

If $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$.
 2. **augmentation**

If $\alpha \rightarrow \beta$, then $\gamma\alpha \rightarrow \gamma\beta$ for any γ .
 3. **transitivity**

If $\alpha \rightarrow \beta$ and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$.
- These rules are
 - **sound** (only generate **correct** functional dependencies) and
 - **complete** (can generate **all** correct functional dependencies).

Closure Example

- $R = \{A, B, C, G, H, I\}$
 $F = \{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$
- Some members of F^+ :
 - $A \rightarrow H$
 - by transitivity from $A \rightarrow B$ and $B \rightarrow H$.
 - $AG \rightarrow I$
 - by augmenting $A \rightarrow C$ with G to get $AG \rightarrow CG$;
 - and by transitivity from $AG \rightarrow CG \rightarrow I$.
 - $CG \rightarrow HI$
 - by augmenting $CG \rightarrow I$ to produce $CG \rightarrow CGI$;
 - then by augmenting $CG \rightarrow H$ to produce $CGI \rightarrow HI$;
 - and by transitivity from $CG \rightarrow CGI \rightarrow HI$.

Pseudo Code for Computing F^+

Algorithm Compute the closure F^+ of F :

```
1.  $F^+ = F$ 
2. repeat
3.   for each functional dependency  $f$  in  $F^+$  do
4.     apply reflexivity and augmentation rules on  $f$ 
5.     add the resulting functional dependencies to  $F^+$ 
6.   end for
7.   for each pair  $f_1$  and  $f_2$  in  $F^+$  do
8.     if  $f_1$  and  $f_2$  can be combined by transitivity then
9.       add the resulting functional dependency to  $F^+$ 
10.    end if
11.  end for
12. until  $F^+$  does not change any further
13. return  $F^+$ 
```

NOTE: We shall see an alternative procedure for this task later.

Closure of Functional Dependencies (Cont.)

- We can further simplify manual computation of F^+ by using the following additional rules:
 1. **union**
if $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$, then $\alpha \rightarrow \gamma\beta$
 2. **decomposition**
if $\alpha \rightarrow \beta\gamma$, then $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$
 3. **pseudotransitivity**
if $\alpha \rightarrow \beta$ and $\gamma\beta \rightarrow \delta$, then $\alpha\gamma \rightarrow \delta$
- These rules can be inferred from Armstrong's axioms.

Closure of Attribute Sets

Given a set of attributes α , define the **closure** of α **under** F (denoted by α^+) as the set of attributes that are functionally determined by α under F .

Algorithm Compute α^+ under F :

1. $\alpha^+ = \alpha$
 2. **repeat**
 3. **for each** functional dependency $\beta \rightarrow \gamma$ in F^+ **do**
 4. **if** $\beta \subseteq \alpha^+$ **then**
 5. $\alpha^+ = \alpha^+ \cup \gamma$
 6. **end if**
 7. **end for**
 8. **until** α^+ does not change any further
 9. **return** α^+
-

NOTE: We shall see an alternative procedure for this task later.

Example of Attribute Set Closure

- $R = \{A, B, C, G, H, I\}$
 $F = \{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$
- Find $\{AG\}^+$
 1. $\{AG\}^+ = \{AG\}$
 2. $\{AG\}^+ = \{ABCG\}$ ($A \rightarrow C$ and $A \rightarrow B$)
 3. $\{AG\}^+ = \{ABCGH\}$ ($CG \rightarrow H$ and $AG \rightarrow ABCG$)
 4. $\{AG\}^+ = \{ABCGHI\}$ ($CG \rightarrow I$ and $AG \rightarrow ABCGH$)
- Is $\{AG\}$ a candidate key?
 1. Is $\{AG\}$ a super key?
 - Does $AG \rightarrow R$? Or does $R \subseteq \{A, G\}^+$?
 2. Is any subset of $\{AG\}$ a superkey?
 - Does $A \rightarrow R$? Or does $R \subseteq \{A\}^+$?
 - Does $G \rightarrow R$? Or does $R \subseteq \{G\}^+$?

Uses of Attribute Closure

There are several uses of the attribute closure algorithm:

- Testing for superkey:
 - To test if α is a superkey, we compute α^+ , and check if α^+ contains all attributes of R .
- Testing functional dependencies:
 - To check if a functional dependency $\alpha \rightarrow \beta$ holds (or, in other words, is in F^+), just check if $\beta \subseteq \alpha^+$.
 - That is, we compute α^+ by using attribute closure, and then check if it contains β .
 - It is a simple and cheap test, and very useful.
- Computing closure of F :
 - For each $\gamma \subseteq R$, we find the closure γ^+ , and for each $S \subseteq \gamma^+$, we output a functional dependency $\gamma \rightarrow S$.

Example

- Let $R = \{A, B, C, D, E, G\}$ and
 $F = \{AB \rightarrow C, AD \rightarrow E, B \rightarrow D, AG \rightarrow B\}$
- Compute $\{A, B\}^+$ and $\{A, G\}^+$

End of Lecture 9